

◆ SECTION 6: NUMERICAL MCQs – RAOULT'S LAW & COLLIGATIVE PROPERTIES (Q151–Q200)

RAOULT'S LAW & VAPOUR PRESSURE (Q151–Q155)

151.

Given:

Pure water vapour pressure = 23.8 mm Hg

1 mol solute + 9 mol water → total moles = 10

$$X_{\text{water}} = \frac{9}{10} = 0.9$$

$$p = X_{\text{water}} p^{\circ} = 0.9 \times 23.8 = 21.42 \text{ mm Hg}$$

✓ Answer: B – 21.4 mm Hg

152.

Relative lowering of vapour pressure:

$$\frac{p^{\circ} - p}{p^{\circ}} = X_{\text{solute}}$$

$$X_{\text{solute}} = \frac{1}{10} = 0.1$$

✓ Answer: A – 0.1

153.

$$X_{\text{solute}} = \frac{2}{2 + 18} = \frac{2}{20} = 0.1$$

✓ Answer: A – 0.1

154.

20% lowering means:

$$\frac{p^{\circ} - p}{p^{\circ}} = 0.20 = X_{\text{solute}}$$

✓ Answer: A – 0.2

155.

$$\Delta T_b = T_{\text{solution}} - T_{\text{pure}} = 374 - 373 = 1 \text{ K}$$

✓ Answer: A – 1 K

BOILING POINT & FREEZING POINT (Q156–Q159)

156.

$$\Delta T_b = K_b m = 0.52 \times 1 = 0.52 \text{ K}$$

✓ Answer: A – 0.52 K

157.

$$\Delta T_f = K_f m = 1.86 \times 1 = 1.86 \text{ K}$$

✓ Answer: A – 1.86 K

158.

$$\Delta T_f = 1.86 \times 0.5 = 0.93 \text{ K}$$

✓ Answer: A – 0.93 K

159.

$$m = \frac{\Delta T_f}{K_f} = \frac{1.86}{1.86} = 1.0$$

✓ Answer: B – 1.0 m

OSMOTIC PRESSURE (Q160–Q162)

Formula:

$$\pi = CRT$$

160.

$$\pi = 0.1 \times 0.082 \times 300 \approx 2.46 \text{ atm}$$

✓ Answer: B – 2.46 atm

161.

$$\pi = 0.2 \times 0.082 \times 300 \approx 4.92 \text{ atm}$$

✓ Answer: B – 4.92 atm

162.

$$\pi \propto C \Rightarrow 0.2M = 2 \times 0.1M$$

✓ Answer: C – 2π

ELECTROLYTES & van't Hoff FACTOR (Q163–Q170)

163.

NaCl dissociates → more particles → higher ΔT_b

✓ Answer: B – NaCl

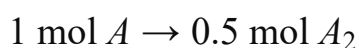
164.

$$m_{\text{effective}} = i \times m = 2 \times 0.5 = 1.0$$

✓ Answer: B – 1.0 m

165.

Dimerization:



✓ Answer: B – 0.5

166.

Observed < expected → fewer particles → association

✓ Answer: A – Association ($i < 1$)

167.

Observed > expected → more particles → dissociation

✓ Answer: B – Dissociation

168.

$$i = \frac{\text{observed}}{\text{calculated}} = 2$$

✓ Answer: C – 2

169.

For $AB \rightarrow A^+ + B^-$:

$$i = 1 + \alpha$$

✓ Answer: A – $1 + \alpha$

170.

For dimerization:

$$i = 1 - \frac{\alpha}{2}$$

✓ Answer: B – $1 - \alpha/2$

ELECTROLYTE BP & FP (Q171–Q172)

171.

$$\Delta T_b = iK_b m = 2 \times 0.52 \times 1 = 1.04$$

Boiling point = $100 + 1.04 \approx 101.04$ °C

✓ Answer: B

172.

$$\Delta T_f = 2 \times 1.86 = 3.72$$

Freezing point = $-3.72\text{ }^{\circ}\text{C}$

✓ Answer: B

RELATIVE LOWERING & MOLE FRACTION (Q173–Q175)

173.

$$\frac{p^{\circ} - p}{p^{\circ}} = X_{\text{solute}}$$

✓ Answer: A

174.

$$X_{\text{solute}} = \frac{0.5}{0.5 + 9.5} = 0.05$$

✓ Answer: C – 0.05

175.

$$X_{\text{solute}} = 1 - 0.98 = 0.02$$

✓ Answer: A – 0.02

OSMOTIC PRESSURE & TONICITY (Q176–Q178)

176.

NaCl dissociates $\rightarrow i \approx 2 \rightarrow$ highest π

✓ Answer: C – 0.1 M NaCl

177.

Same π at same T $\rightarrow$ isotonic

✓ Answer: D

178.

Reverse osmosis needs **pressure > osmotic pressure**

✓ Answer: C

APPLICATIONS & CONCEPTS (Q179–Q185)

179.

Most sensitive for polymers → osmotic pressure

✓ Answer: C

180.

Same ΔT_f :

$$\Delta T_f \propto \frac{1}{M}$$

If 1 g A $\equiv$ 2 g B → A has **half molar mass**

✓ Answer: B

181.

Lowering of vapour pressure → molar mass

✓ Answer: B

182.

$i = 1.5$ → partial dissociation (1 → 2 type)

✓ Answer: B

183.

$$i = \frac{0.372}{0.186} = 2$$

✓ Answer: B

184.

Works at room T, very dilute solutions

✓ Answer: A

185.

Semipermeable membrane → solvent only

✓ Answer: B

OSMOTIC PRESSURE COMPARISONS (Q186–Q188)

186.

1 M vs 0.5 M → first has double π

✓ Answer: C

187.

No net osmosis → isotonic

✓ Answer: C

188.

$$\Delta T_f = 0.93 = 1.86m \Rightarrow m = 0.5$$

$$\Delta T_b = 0.52 \times 0.5 = 0.26$$

✓ Answer: A

GENERAL COLLIGATIVE CONCLUSIONS (Q189–Q200)

189.

Non-volatile solute:

- BP ↑
- FP ↓

✓ Answer: B

190.

$\text{Na}_2\text{SO}_4 \rightarrow 3 \text{ ions}$

✓ Answer: B

191.

Association $\rightarrow$ fewer particles

✓ Answer: B

192.

Dissociation $\rightarrow$ more particles

✓ Answer: A

193.

Same molality + non-electrolytes $\rightarrow$ same effect

✓ Answer: B

194.

Non-volatile solute $\rightarrow$ BP increases

✓ Answer: B

195.

Depends on number of solute particles

✓ Answer: A

196.

$$\pi \propto C$$

✓ Answer: A – Doubles

197.

Greater $\Delta T_f \rightarrow$ more particles

✓ Answer: B

198.

$$\frac{300 - 270}{300} = 0.1$$

✓ Answer: A

199.

Main use $\rightarrow$ molar mass determination

✓ Answer: B

200.

Total questions = 200

✓ Answer: C